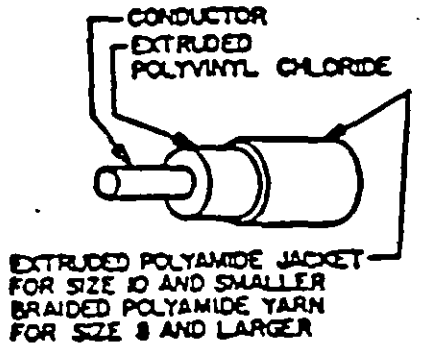
INCH-POUND
MIL-W-16878/19A(NAVY)
11 September 1992
SUPERSEDING
MIL-W-16878/19(NAVY)
10 August 1981

MILITARY SPECIFICATION SHEET

WIRE, ELECTRICAL, POLYVINYL CHLORIDE (PVC) INSULATED, 105°C, 3000 VOLTS, POLYAMIDE COVERING (NOT FOR NAVY SHIPBOARD USE)

This specification is approved for use by the Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-W-16878.



(NOT FOR NAVY SHIPBOARD USE)

FIGURE 1. Construction.

AMSC N/A FSC 6145

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:30Z
Check the source to verify that this is the current version before use.

TABLE I. Construction details.

(See footnotes at the end of table)

PIN 3/	Wire size	Stranding	Conductor		Conductor diameter (nominal) (inch)	Finished wire diameter (inch)		Polyamide thickness (inch)	
			Material 1/ 2/	Coating		Min	Max	Min	Max
M16878/19-BEA*	24	1 X 24	Copper	Tin	0.0201	0.076	0.095	0.0025	0.0045
M16878/19-DEA*	24	1 X 24	H.S.C.A.	Silver	.0201	.076	.095	.0025	.0045
M16878/19-BEB*	24	7 X 32	Copper	Tin	.024	.080	.099	.0025	.0045
M16878/19-DEB*	24	7 X 32	H.S.C.A.	Silver	.024	.080	.099	.0025	.0045
M16878/19-BEE*	24	19 X 36	Copper	Tin	.026	.080	.099	.0025	.0045
M16878/19-DEE*	24	19 X 36	H.S.C.A.	Silver	.026	.080	.099	.0025	.0045
M16878/19-BFA*	22	1 X 22	Copper	Tin	.0254	.081	.101	.0025	.0045
M16878/19-DFA*	22	1 X 22	H.S.C.A.	Silver	.0254	.081	.101	.0025	.0045
M16878/19-BFB*	22	7 X 30	Copper	Tin	.030	.086	.105	.0025	.0045
M16878/19-DFB*	22	7 X 30	H.S.C.A.	Silver	.030	.086	.105	.0025	.0045
M16878/19-BFE*	22	19 X 34	Copper	Tin	.032	.088	.105	.0025	.0045
M16878/19-DFE*	22	19 X 34	H.S.C.A.	Silver	.032	.088	.105	.0025	.0045
M16878/19-BGA*	20	1 X 20	Copper	Tin	.0320	.088	.107	.0025	.0045
M16878/19-DGA*	20	1 X 20	H.S.C.A.	Silver	.0320	.088	.107	.0025	.0045
M16878/19-CGA*	20	1 X 20	C.C. steel	Tin	.0320	.088	.107	.0025	.0045
M16878/19-BGB*	20	7 X 28	Copper	Tin	.038	.094	.113	.0025	.0045
M16878/19-DGB*	20	7 X 28	H.S.C.A.	Silver	.038	.094	.113	.0025	.0045
M16878/19-BGE*	20	19 X 32	Copper	Tin	.041	.094	.113	.0025	.0045
M16878/19-DGE*	20	19 X 32	H.S.C.A.	Silver	.041	.094	.113	.0025	.0045
M16878/19-BHA*	18	1 X 18	Copper	Tin	.0403	.096	.116	.0025	.0045
M16878/19-BHB*	18	7 X 26	Copper	Tin	.049	.105	.124	.0025	.0045
M16878/19-BHE*	18	19 X 30	Copper	Tin	.051	.105	.124	.0025	.0045
M16878/19-BJA*	16	1 X 16	Copper	Tin	.0508	.108	.129	.003	.006
M16878/19-BJE*	16	19 X 29	Copper	Tin	.059	.116	.137	.003	.006
M16878/19-BJF*	16	26 X 30	Copper	Tin	.062	.119	.140	.003	.006
M16878/19-BKA*	14	1 X 14	Copper	Tin	.0641	.121	.142	.003	.006
M16878/19-BKE*	14	19 X 27	Copper	Tin	.072	.129	.150	.003	.006
M16878/19-BKH*	14	41 X 30	Copper	Tin	.080	.137	.158	.003	.006

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:30Z
Check the source to verify that this is the current version before use.

PIN 3/	Wire size	Stranding	Conductor		Conductor diameter (nominal) (inch)	Finished wire diameter (inch)		Polyamide thickness (inch)	
			Material 1/ 2/	Coating		Min	Max	Min	Max
M16878/19-BLA*	12	1 X 12	Copper	Tin	0.0808	0.159	0.171	0.003	0.006
M16878/19-BLE*	12	19 X 25	Copper	Tin	.092	.162	.183	.003	.006
M16878/19-BLG*	12	37 X 28	Copper	Tin	.089	.159	.180	.003	.006
M16878/19-BLJ*	12	65 X 30	Copper	Tin	.098	.168	.189	.003	.006
M16878/19-BMA*	10	1 X 10	Copper	Tin	.1019	.174	.197	.004	.008
M16878/19-BMG*	10	37 X 26	Copper	Tin	.111	.183	.206	.004	.008
M16878/19-BMK*	10	105 X 30	Copper	Tin	.127	.199	.222	.004	.008
M16878/19-BNL*	8	133 X 29	Copper	Tin	.169	.253	.280	.006	.012
M16878/19-BPL*	6	133 X 27	Copper	Tin	.213	.300	.329	.007	.014
M16878/19-BRL*	4	133 X 25	Copper	Tin	.268	.362	.396	.007	.014
M16878/19-BSP*	2	665 X 30	Copper	Tin	.342	.436	.470	.007	.014
M16878/19-BTR*	1	817 X 30	Copper	Tin	.382	.486	.520	.007	.014
M16878/19-BUS*	0	1045 X 30	Copper	Tin	.431	.525	.545	.007	.014

1/ H.S.C.A. stands for high-strength copper alloy.
2/ C.C. stands for copper-clad.
3/ PIN stands for part or identifying number.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:30Z
Check the source to verify that this is the current version before use.

ADDITIONAL REQUIREMENTS:

Visual and mechanical examination: Required.
Spark test: 8.0 kV.
Impulse dielectric test: 12.0 kV.
Dielectric withstanding voltage: 6.0 kV.
Insulation resistance: $IR = K \log_{10} D/d$.

Where: IR = Minimum insulation resistance in megohms per 1000 feet at 20°C.
K = 2000.
D = Maximum average diameter of finished wire.
d = Conductor diameter.

Cold bend: Condition 4 hours at minus 54 ± 1°C (see table II).

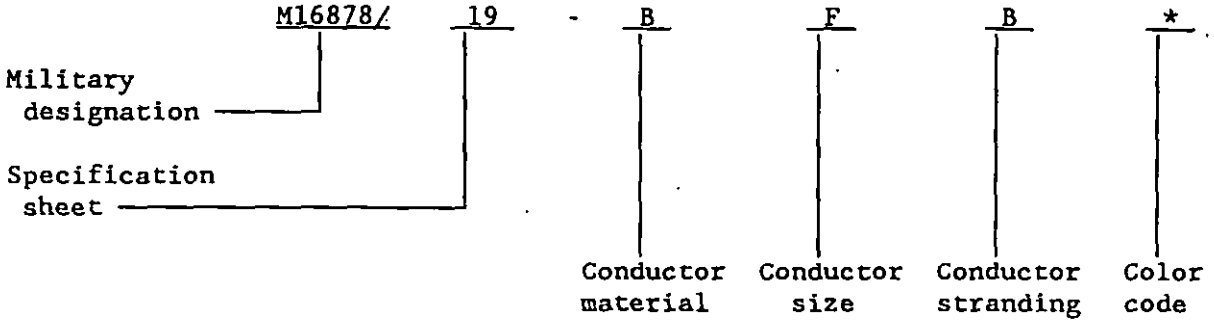
TABLE II. Cold bend mandrel sizes.

Wire size	Cold bend mandrel (maximum diameter in inches)
24, 22	2
20 through 14	3
12, 10	4.5
8	6
6, 4	10
2, 1, 0	18

Concentricity: 70 percent (minimum).
Surface resistance: 5 megohms (minimum) for braided jackets on size 8 or larger.
Wrap back: Not required.
Heat resistance: Condition at 150°C. Cracking of the jacket shall constitute a failure.
Flammability: Meets requirements of UL VW1 Flame Test.
Toxicity: Meets requirements of concentration levels specified in MIL-E-917.
Shrinkage: Required.
Heat aging: 25 percent change (maximum) in 96 hours at 136°C.
Insulation tensile strength: 1800 pounds force per square inch (minimum).
Insulation elongation: 100 percent (minimum).
Marking and stripe durability: Required.
Fungus resistance: Required.
Maximum dc resistance of finished wire: See table I of MIL-W-16878.
Identification of product: Required for wire sizes 22 AWG and larger.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:30Z
Check the source to verify that this is the current version before use.

PIN example (see MIL-W-16878):



Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 6145-N330-18)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:30Z
Check the source to verify that this is the current version before use.